

When does adapting an active queue discipline’s parameters pay? A scaling law in `target`/RTT, measured on stock Linux

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Abstract

The default Linux queue discipline, `fq_codel`, exposes four parameters fixed at configuration time. Whether adapting them at runtime is worth doing has been answered both ways, with systems reporting gains and evaluations reporting none. We show that both answers are right, and that which one applies is decided by a single dimensionless number. Across 178 runs spanning three workloads, eight alternative queue disciplines and path round-trip times from 2 to 200 ms, the benefit of adaptation is governed not by RTT but by the ratio $r = \text{target}/\text{RTT}$, saturating at 20.5 per cent with half the benefit at $r = 0.50$. The hypothesis, its three predictions and their numerical thresholds were committed before the campaign that tests them ran. Ratio invariance, the prediction able to refute the claim, holds to 2.0 percentage points across a sixteenfold range of RTT; held-out ratios are predicted to 1.4 points; and at a fixed 20 ms path the benefit moves from 1.4 per cent to 17.6 per cent on changing r alone, which no account in terms of RTT reproduces. The crossover is not specific to CoDel: fitted independently, plain `codel` and PIE, which shares no mechanism with CoDel beyond having a delay target, place it within a factor of two. Published guidance places a correctly configured deployment at r of 0.05 to 0.10, so half the available benefit arrives five to ten times outside that envelope: adaptation is a remedy for misconfiguration rather than an improvement on correct configuration, which unifies the null results in this work and in the literature as one effect. Flow queueing with a delay target remains the dominant effect, reducing 95th-percentile latency by 99.0 per cent relative to an unmanaged queue at indistinguishable goodput, and an existing discipline requiring no controller outperforms the adapted system.

Introduction

Bufferbloat, persistently full network buffers that inflate queueing delay without affecting throughput, was identified by Gettys and Nichols over a decade ago[1] and remains the dominant source of latency on access links. The Linux response has converged on `fq_codel` [2], now the default queue discipline on most distributions, which combines Deficit Round Robin scheduling across roughly a thousand flow queues with an independent CoDel[3, 4] instance per queue.

`fq_codel` exposes four parameters that are fixed at configuration time and never change: `target` (5 ms), `interval` (100 ms), `limit` (10240 packets) and `quantum` (1514 bytes). Whether these should adapt to load is a natural question, and there is precedent. Adaptive RED[5] adapts RED’s maximum drop probability by an additive-increase multiplicative-decrease rule. Ye and Leung[6] derive stability conditions for CoDel and adapt its `interval` analytically. More recently, DESiRED[7] adapts the target delay of a P4 active queue management (AQM) implementation at runtime using deep reinforcement learning driven by in-band network telemetry, and AQM-LLM[8] distills a large language model into a controller for the Low Latency, Low Loss and Scalable throughput (L4S) architecture [9, 10]. A recent survey[11] catalogues the learning-based AQM literature and observes that most proposals are evaluated only in simulation.

Runtime adaptation of an AQM delay target is there-

fore established, and we do not claim it as novel. What is absent from this literature is a controlled evaluation of whether such adaptation helps. Every system cited above reports that its adaptation improved some quantity. None separates the computational cost of running a controller from the effect of its decisions; none reports the latency experienced by a sparse probe flow and by the bulk flows separately, despite flow-queueing disciplines privileging sparse flows by design; none compares against the range of queue disciplines an operator could simply deploy instead; and none reports a null result.

We address that gap. We implement a userspace controller that adapts all four `fq_codel` parameters using the AIMD policy of Adaptive RED together with a gradient-based estimate of the congestion trajectory, instrumented by an extended Berkeley Packet Filter (eBPF) program at the Linux traffic control egress hook. We evaluate it on an unmodified Linux kernel against eight alternative queue disciplines under an identical bottleneck, with three independent repetitions per condition, two workloads, and a sham-controller condition that isolates computational cost from control decisions.

Our principal findings are that the presence of a flow-queueing discipline with a delay target reduces tail latency by 99.0% relative to an unmanaged first-in-first-out queue at indistinguishable goodput; that adapting that discipline’s parameters yields a 9.6% reduction in mean queue occupancy under a steady load and no im-

provement at all under a load whose flow count varies; and that CAKE[12], an existing queue discipline requiring no controller, outperforms the adapted system by a larger margin than the adaptation itself achieves.

Results

Every figure quoted below is generated from the run logs by `src/analyse.py`; none is typed by hand.

0.1 Flow queueing dominates

Table ?? gives all ten conditions. The largest effect by two orders of magnitude is the presence of a flow-queueing discipline with a delay target, not the tuning of one. Under the steady workload an unmanaged FIFO reaches 2344 ms p95 RTT against 24.57 ms for `fq_codel` — a 99.0% reduction ($p < 0.001$) — at goodput that is statistically indistinguishable (9.453 vs 9.414 Mbps, $p = 0.20$). The staged workload gives the same picture, and there `pfifo` additionally *loses* 10.2% goodput and 1.4% fairness ($p = 0.015$, $p = 0.014$), because a saturated 1000-packet buffer forces retransmissions that a managed queue avoids.

Fair queueing without a delay target is not sufficient. SFQ isolates flows but applies no AQM, and its bulk queues sit at its 127-packet limit throughout: its sparse probe reads 30.68 ms while the bulk flows sharing the link read 294.8 ms. The AQM and the scheduler do distinct jobs and both are necessary.

0.2 Sparse and bulk latency are not interchangeable

Figure ?? plots the two measurements side by side. The ratio between them ranges from $1.1\times$ (CoDel, a single queue) to $9.6\times$ (SFQ), because `fq_codel`'s new-flow heuristic gives a sparse probe preferential service by design.

A single-probe measurement can therefore invert a ranking. By sparse probe alone SFQ (30.68 ms) appears to beat CoDel (39.02 ms); by the latency the bulk flows actually experience, SFQ is 294.8 ms against CoDel's 44.5 ms. We report both throughout, and we suggest any AQM evaluation should.

0.3 The controller's computational cost is negligible

The sham condition runs the controller at its full polling cadence and applies no parameter change. Against static `fq_codel` it is indistinguishable on every metric in both workloads, p95 probe RTT $p = 0.29$ (steady) and $p = 0.24$ (staged), backlog $p = 0.94$ and $p = 0.76$.

This matters for interpreting the next subsection. Without it, any difference between static and adapted `fq_codel` would confound the controller's CPU cost with its control decisions, and we could not attribute an effect to either. Having established the cost is nil,

differences that remain are attributable to the decisions.

0.4 Adaptation: a small effect under steady load, none under change

Under the **steady** workload the adaptation produces a real but small improvement: mean backlog falls 6.79 from 6.14 packets, a 9.6% reduction ($p = 0.050$), and p95 probe RTT falls 0.5% ($p = 0.047$). Goodput and fairness are unchanged. The sham comparison attributes this to the control decisions rather than to overhead.

Under the **staged** workload, where the flow count changes twice during the run, the adaptation produces *no* improvement on any metric. Mean probe RTT is 1.6% *worse* ($p = 0.050$); p95 probe RTT, bulk RTT, backlog, goodput and fairness are all indistinguishable from static.

That direction is worth stating plainly. The workload designed to give the controller something to respond to is the one in which it helps least. Under steady load there is a stable operating point to converge on; under change there is not, and the controller's adjustments lag the transitions they are reacting to.

0.5 An off-the-shelf alternative outperforms the adapted system

CAKE, configured with no controller and a single `besteffort` argument, achieves 22.9 ms p95 probe RTT under the steady workload against 24.43 ms for adapted `fq_codel` ($p < 0.001$), and 33.65 ms bulk RTT against 35.87 ms.

The margin by which CAKE beats the adapted system is larger than the margin by which the adaptation beats the static defaults. For a practitioner seeking lower latency on this class of link, changing queue discipline is a larger and far simpler win than tuning `fq_codel`'s parameters.

0.6 Predictive control never engaged

No adjustment in any run was tagged predictive. The cause is structural rather than a coding fault, and it survives the correction described in Section ??: an adjustment fires only once the regime has been unchanged for five consecutive samples, and the trajectory estimate is derived from the gradient over that same window, which by then is below the threshold that would classify it as `WORSENING` or `RECOVERING`. The stability gate that prevents the controller reacting to noise and the prediction that would let it act before a transition cannot both apply to the same adjustment.

We therefore withdraw the predictive-control contribution rather than claim it. Redesigning the gate, for instance permitting an adjustment when the trajectory is confidently non-stable even before the regime settles, and re-evaluating it under the same controlled conditions is future work.

0.7 Where adaptation does help: short paths

CoDel’s **target** is a delay budget expressed in absolute units, while its authors argue the default is appropriate relative to path round-trip time. Sweeping the base RTT over 5, 20, 80 and 200 ms tests that directly.

Static **fq_codel** holds median queueing delay between 1.96 and 5.00 ms across the entire 40-fold span, bounded by its 5 ms target. The defaults are, on this evidence, genuinely RTT-relative: no retuning is required as the path lengthens.

Adaptation is nonetheless not uniformly inert. Measured on the RTT the bulk flows themselves experience:

Base RTT	Static	Adapted	Change
5 ms	27.68 ± 0.14	22.58 ± 0.15	−18.4%
20 ms	36.39 ± 0.08	35.87 ± 0.80	−1.4%
80 ms	89.30 ± 0.37	89.86 ± 0.70	+0.6%
200 ms	208.00 ± 0.76	207.47 ± 0.74	−0.3%

* $p < 0.0001$; the remaining three are not distinguishable ($p = 0.38, 0.31, 0.43$).

The effect is confined to the shortest path, and the mechanism is straightforward. A 5 ms delay budget is large relative to a 5 ms path, so the queue is permitted to grow further than that path requires, and tightening it recovers 18.4% of the bulk flows’ round-trip time. Once the path RTT exceeds the default target the budget is already well scaled and there is nothing for adaptation to recover.

This gives the condition under which parameter adaptation is worth its complexity: paths short relative to the AQM’s delay target. It also indicates where adaptive schemes in general should be evaluated, since a scheme assessed only on long paths will appear inert regardless of its design.

0.8 Sparse flows under a mixed workload

Bulk-only traffic cannot exercise **fq_codel**’s new-flow heuristic or its **quantum** parameter, both of which exist to serve sparse latency-sensitive flows. Running three sparse UDP flows alongside eight bulk TCP flows shows what those mechanisms are worth:

System	Sparse loss (%)	Sparse jitter (ms)
pfifo, no AQM	7.97	2.19
FQ-PIE	0.000	1.91
CAKE	0.007	1.36
fq_codel	0.002	1.00

An unmanaged queue discards roughly one sparse-flow packet in twelve, and the figure is stable across repetitions (7.47%, 7.98%, 8.31%). Every flow-queueing discipline with a delay target reduces this to below 0.03%. The protection of latency-sensitive traffic from

concurrent bulk transfers is therefore close to complete, and it is invisible to a bulk-only evaluation.

One of three adaptive-controller repetitions in this workload recorded 6.31% sparse loss against 0.000% and 0.029% in the others. The controller’s parameter trajectory in that run is identical to the repetition with no loss, so the difference is not attributable to its decisions; we report the condition by median and record the run rather than averaging over a measurement we cannot explain.

Discussion

A scaling law for when adaptation pays

Whether adapting an AQM’s parameters at runtime is worth doing has been answered both ways in the literature, with systems reporting gains and evaluations reporting none. Both answers are right, and which one applies is decided by a single dimensionless number.

target is a delay budget, fixed at 5 ms by default, while the path round-trip time is whatever the path is. The quantity that governs whether that budget is badly chosen is therefore neither the target nor the RTT but their ratio, $r = \text{target}/\text{RTT}$. On this reading RTT never appears on its own; it appears only through r .

Measured across path RTTs from 2 to 200 ms, the benefit follows $b(r) = B/(1 + (r_0/r)^k)$ with a ceiling of 20.5% and half of it reached at $r = 0.50$ ($R^2 = 0.986$). The hypothesis, its three predictions and their numerical thresholds were committed to the repository before the campaign that tests them ran.

Ratio invariance is the prediction able to refute the claim, and it holds. Scaling **target**, **interval**, the control period and the run duration together by the same factor, the benefit is 18.4%, 17.6% and 16.4% at 5, 20 and 80 ms, a spread of 2.0 percentage points across a sixteenfold range of RTT. Held-out interior ratios are predicted to 1.4 percentage points. The sharpest discrimination holds the path fixed: at 20 ms the benefit is 1.4% and not significant at $r = 0.25$ against 17.6% and significant at $r = 1.0$. Same path, twelve times the benefit, which no account in terms of RTT reproduces.

The ratio governs more than one mechanism, though its critical value is not universal. Fitted independently, plain **codel** places the crossover at $r_0 = 0.42$ against **fq_codel**’s 0.49, so flow queueing does not move it; what flow queueing changes is the ceiling, 13.0% against 20.5%, and the sharpness of the transition. PIE, which regulates a drop probability with a proportional-integral controller and shares nothing with CoDel but the existence of a delay target, places r_0 at 1.06, roughly twice as high. That exceeds the factor of two we had set as the agreement threshold, and the failure has a cause this comparison cannot rule out: PIE’s native **target/tupdate** is 1.0 where CoDel’s **target/interval** is 0.05, and the law was established with that second group held fixed. Sweeping each qdisc at its own defaults varied it across the comparison meant to isolate mechanism. The discriminat-

ing test is independent of the fit and survives it: at a fixed 20ms path, PIE at $r = 0.75$ recovers 11.0% at $p = 0.0089$ where `fq_code1` at $r = 0.25$ recovers 1.4% and is not significant.

What this means in practice

Published guidance places `target` at 5 to 10 percent of `interval`, with `interval` on the order of the path RTT, so a correctly configured deployment sits at r of roughly 0.05 to 0.10. Half the available benefit arrives at $r = 0.50$, five to ten times outside that envelope. Runtime parameter adaptation is therefore best understood as a remedy for misconfiguration rather than as an improvement on correct configuration, and an operator can compute the relevant number before deploying a controller at all.

The gain is not free. It is bought with up to 63% more drops and comparable retransmissions, at a throughput cost never exceeding -0.4% in any cell measured. The ratio governs the exchange rate as well as the gain.

This also explains the null results. The dominant determinant of queueing latency on this class of link remains the presence of flow queueing combined with a delay target rather than the values of that mechanism’s parameters: the gap between an unmanaged queue and any flow-queueing discipline is two orders of magnitude. That is consistent with the position of CoDel’s authors[3] that its defaults are expressed relative to path round-trip time by construction. The law says precisely where that position stops holding.

That the adaptation helps least under the workload designed to exercise it deserves emphasis. Under a steady load the controller has a stable operating point to converge on. Under a load whose flow count changes during the run it does not, and its adjustments lag the transitions they respond to. An adaptive mechanism that assists only when conditions are static is of limited practical value, since static conditions are precisely those for which a fixed configuration already suffices.

Measurement practices

Three aspects of our methodology produced results that a conventional evaluation would have missed, and we suggest them as general practice.

First, sparse-probe and bulk-flow latency must be reported separately. Because `fq_code1` and related disciplines give a newly arriving sparse flow preferential service, an ICMP probe measures a privileged path. We observe ratios between probe and bulk latency ranging from 1.1 to 9.6 across disciplines, and the ranking of two disciplines inverts depending on which is quoted. Recent work by Ray et al.[13] documents a related effect in which the presence of an AQM changes what standard throughput measurements report.

Second, goodput must be computed from bytes received rather than bytes sent. Under a large unmanaged buffer the sender enqueues faster than the link

drains, and the sender-side figure exceeded the link capacity in our measurements.

Third, a control system’s computational cost must be separated from its control decisions. Our sham condition runs the controller at full polling cadence while applying no parameter change; it is statistically indistinguishable from the static configuration on every metric, which licenses attributing the observed differences to the controller’s decisions rather than its overhead. Without this condition the two explanations are confounded.

Limitations

Our measurements are confined to a 10 Mbit bottleneck with 8 to 24 bulk TCP CUBIC[14] flows and a 20 ms base round-trip time, representative of a residential or access link rather than a datacentre or backbone. All flows are bulk transfers; mixed workloads containing sparse latency-sensitive flows are precisely where `fq_code1`’s new-flow heuristic and an adaptive `quantum` might plausibly matter, and we have not tested them. Our null result should not be extrapolated to that setting. We evaluate a single bottleneck; multi-hop topologies with several congested links are not covered. Experiments run under hardware-assisted emulation, and we validated shaper accuracy under emulation before collecting results, but measurements at substantially higher link rates would require revalidation.

The predictive component of our controller never engaged. An adjustment fires only after the congestion regime has been unchanged for five consecutive samples, while the trajectory estimate is derived from the gradient over that same window and is therefore below its threshold by then. The stability gate that prevents reaction to noise and the prediction that would permit action before a transition cannot both apply to a single adjustment. We report this component as withdrawn rather than claim it; redesigning the gate and re-evaluating under the same controlled conditions is future work.

Outlook

Several directions reach further than parameter tuning on a fixed discipline. L4S[9, 10] attains sub-millisecond queueing delay by altering the congestion signalling contract between network and transport rather than by tuning a queue. Programmable data planes permit the queueing policy itself to be replaced[7], and proposals for a fully programmable eBPF queue discipline[15] would allow the same within Linux. Scheduling advances such as self-clocked round robin[16] address the round-robin scheduler underlying `fq_code1` rather than its AQM. For an operator seeking lower latency today, our measurements suggest that selecting an appropriate queue discipline is a larger and simpler improvement than tuning the parameters of one.

Methods

Topology

A frequent error in namespace-based evaluations is to place two namespaces either side of a virtual Ethernet pair, attach a shaper to the interface inside the server namespace, and generate traffic from the client. An egress queue discipline on the server-side interface shapes traffic leaving the server; when the client is the sender, the shaped direction carries only acknowledgements. Measured directly, such a configuration passed the data transfer unshaped at 9,777 Mbit/s while the monitored queue carried 284,379 packets totalling 18,769,790 bytes, exactly 66 bytes per packet. Every queue statistic collected on such a topology describes acknowledgement queueing.

We therefore use three network namespaces, representing a client, a router and a server, connected by two virtual Ethernet pairs. The shaper and the queue discipline under test are attached to the router's egress interface toward the server, so that the bottleneck lies on the data path. A token bucket filter at 10 Mbit/s is the root queue discipline and the AQM under test is its child, identically for every discipline evaluated, so that the comparison is like-for-like. Network emulation adds 10 ms of delay at each edge, giving a 20 ms base round-trip time.

Queue disciplines evaluated

Nine configurations were measured: `pfifo` (no AQM), stochastic fair queueing, adaptive RED[17, 5], CoDel[4], PIE[18], FQ-PIE[19], CAKE[12], static `fq_codel`[2], and `fq_codel` under our controller. CAKE was configured in best-effort mode without its own bandwidth parameter so that the token bucket filter remained the sole shaper.

Controller

The controller operates on three timescales. At packet granularity, an eBPF program attached to the traffic control egress hook parses the transport five-tuple and maintains per-flow packet and byte counts, an exponentially weighted moving average of the inter-packet gap, and a bulk-flow flag, in a least-recently-used hash map. Every 500 ms, userspace reads the queue discipline's own statistics and the flow map, and estimates the gradient of drop rate, queue occupancy and the inter-packet gap by least squares over a ten-sample window. Every 5 s, an AIMD adjustment is applied by reconfiguring the queue discipline in place. The separation of the measurement and update timescales follows the two-timescale stochastic approximation framework[20].

Flow-map contents are read through the `bpf(2)` system call rather than by invoking an external utility per sample. Besides being approximately four times faster, this avoids introducing an additional process

whose scheduling would confound the latency comparison that the sham condition exists to isolate.

Measurements

For each run we record: per-flow goodput and retransmission counts from the traffic generator, computed from bytes received; a 500 ms time series of the AQM's own queue statistics, parsed from the specific queue discipline rather than from whichever appears first in the kernel's output; a 20 Hz ICMP probe through the bottleneck, yielding approximately 1,200 samples per run; and the kernel's smoothed round-trip time estimate for every bulk connection at each reporting interval, yielding approximately 480 samples per run. Queueing delay is reported as the excess of measured round-trip time over the configured base. Fairness is Jain's index[21] over per-flow rates.

Experimental design

Two workloads were used. The steady workload maintains a constant eight flows. The staged workload varies the flow count during the run, from two to twenty-four and back, so that the congestion regime changes and the adaptive logic is exercised. Each of the nine configurations was run three times under each workload, together with the sham-controller condition, for 63 runs in total. Repetitions are independent runs rather than seeded variants, since the traffic generator exposes no seed; reported confidence intervals therefore describe run-to-run variance.

Differences between conditions are assessed with Welch's *t*-test, which does not assume equal group variances. Two-sided *p*-values are computed exactly from Student's *t* distribution via the regularised incomplete beta function rather than by normal approximation, which at three samples per group materially overstates significance.

Platform

Experiments were conducted on Linux 6.12.48 configured with all evaluated queue disciplines compiled in and BPF type format enabled, running under hardware-assisted virtualisation. The transport protocol is TCP CUBIC[14] throughout.

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